Topic 6 Integration

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6.1 The Definite Integral

6.1 The Definite Integral

Definition 6.1

The definite integral of a function f from a to b, denoted

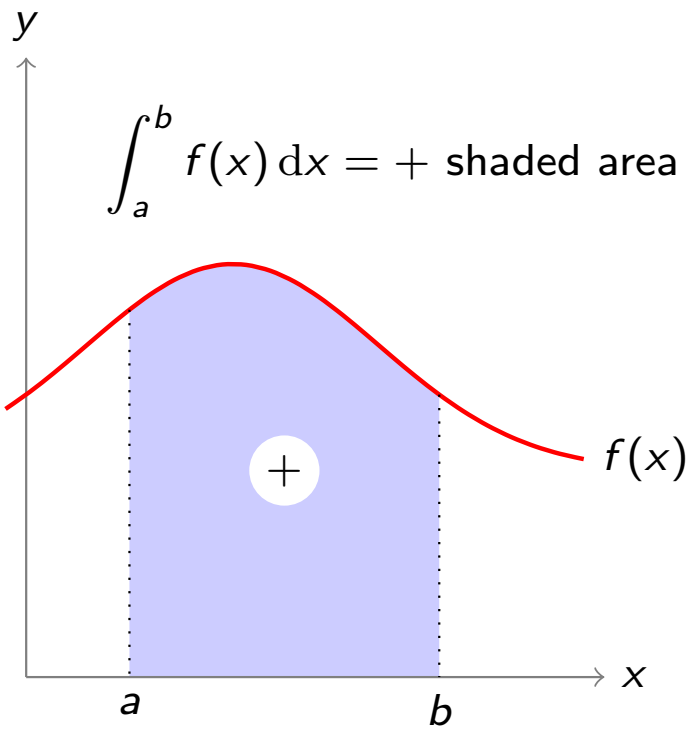
$$\int_a^b f(x)\,\mathrm{d}x,$$

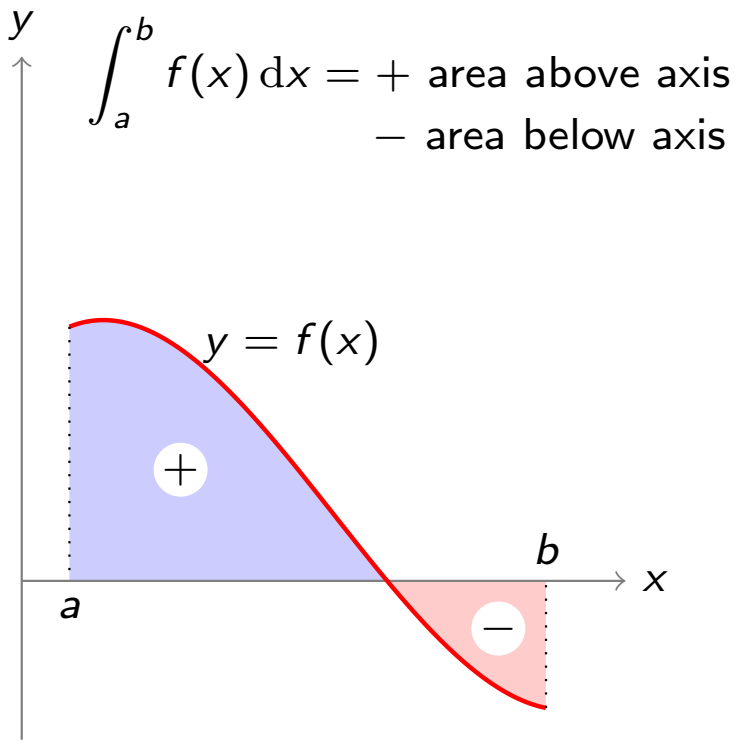
is defined to be the sum of the signed areas bounded by:

- the curve $y = f(x)$,
- the x-axis,
- the vertical lines $x = a$ and $x = b$.

- The values a and b are called the terminals of the integral and $[a, b]$ is called the interval of integration.
- The function $f(x)$ that we are integrating is called the integrand.

6.1 The Definite Integral





6.1 The Definite Integral

6.1 The Definite Integral: Properties

Theorem 6.2

Let f and g be integrable functions.

- $$\int_a^b f(x) + g(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_a^b g(x)\,\mathrm{d}x;$$
- $$\int_a^b k\,f(x)\,\mathrm{d}x = k\int_a^b f(x)\,\mathrm{d}x;$$

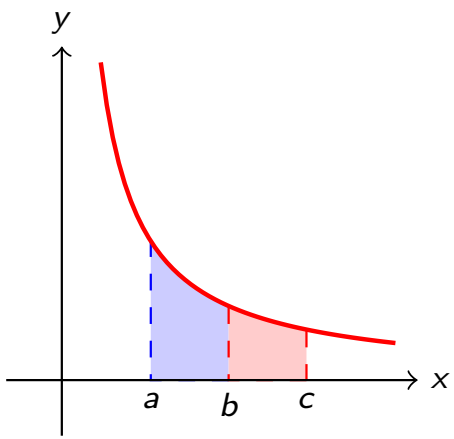
Aside: Linear Functions

6.1 The Definite Integral: Properties

Theorem 6.3

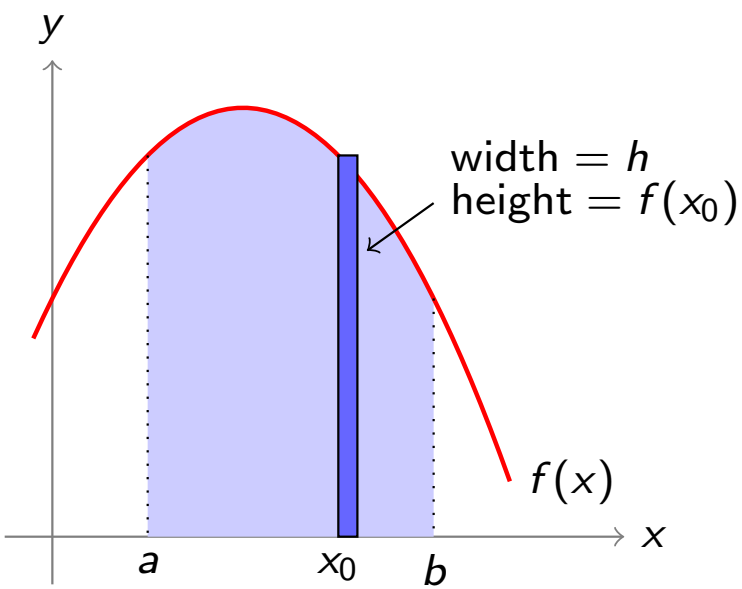
Let f be an integrable function.

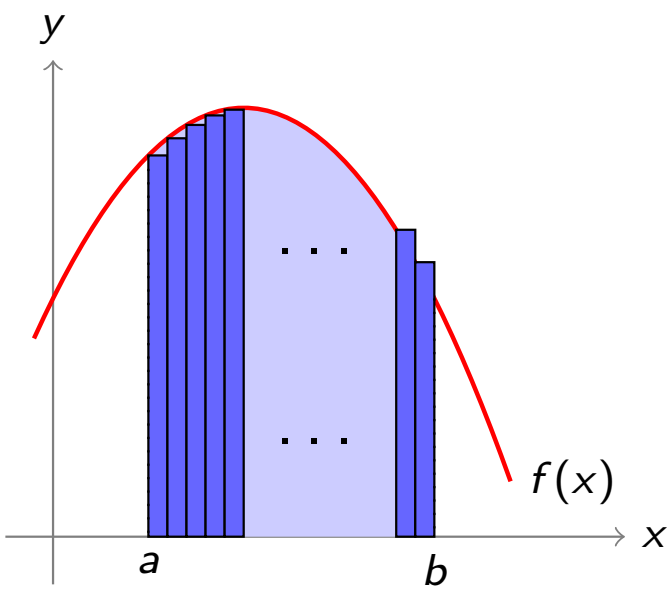
- $\displaystyle\int_a^a f(x)\,\mathrm{d}x = 0;$
- $\displaystyle\int_a^c f(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_b^c f(x)\,\mathrm{d}x$
- $\displaystyle\int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x.$



6.1 The Definite Integral: Computing Integrals Geometrically

- We can approximate the area under the graph by vertical strips.
- This means dividing the area into thin vertical rectangles of equal width.
- As we make the width smaller, the approximation improves.





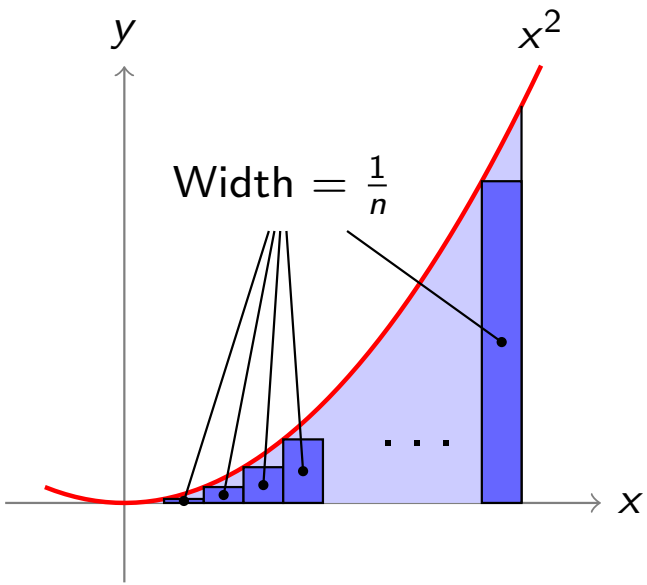
6.1 The Definite Integral: First Principles

Example 6.4

Use the fact that for all $n > 1$:

$$1^2 + 2^2 + 3^2 + \cdots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}$$

to evaluate $\displaystyle\int_0^1 x^2 \, \mathrm{d}x$.



6.1 The Definite Integral: First Principles

6.1 The Definite Integral: Fundamental Theorem

- Surprisingly there is a link between integration and differentiation.
- In a sense, integration reverses differentiation (and vice versa)

Definition 6.5

Let $f : I \longrightarrow \mathbb{R}$. We say a function F is an <u>antiderivative</u> of f when F is differentiable on I and

$$F'(x) = f(x) \qquad \text{for all } x \in I.$$

6.1 The Definite Integral: Fundamental Theorem

Theorem 6.6 (Fundamental Theorem of Calculus)

If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b F'(x)\,\mathrm{d}x = F(b) - F(a) = [F(x)]_a^b\,.$$

6.1 The Definite Integral: First Principles

Example 6.7

Use the fact that $F(x) = \frac{1}{3}x^3$ is an antiderivative of x^2 to evaluate

$$\int_0^1 x^2 \, \mathrm{d}x.$$

6.1 The Definite Integral: Uniqueness

- Antiderivatives are not unique. For any F that has the property
$$F'(x) = f(x),$$
so does $F(x) + C$ for any constant C:
$$\frac{d}{dx}[F(x) + C] = F'(x) = f(x).$$
- We often write the operation of anti-differentiation using an integral sign $\int$ without terminals.
- This is also called the <u>indefinite integral</u>. We write:
$$\int f(x)\,\mathrm{d}x = F(x) + C,$$
to mean "every antiderivative of $f(x)$ is of the form $F(x) + C$, where F is a specific antiderivative of $F(x)$".

6.1 The Definite Integral: Antiderivatives

$$\int x^a \, \mathrm{d}x = \frac{x^{a+1}}{a+1} + C \qquad \text{for } a \neq -1$$

$$\int \frac{1}{x} \, \mathrm{d}x = \int x^{-1} \, \mathrm{d}x = \log(|x|) + C$$

$$\int e^x \, \mathrm{d}x = e^x + C$$

$$\int \sin(x) \, \mathrm{d}x = -\cos(x) + C$$

$$\int \cos(x) \, \mathrm{d}x = \sin(x) + C$$

$$\int \sec^2(x) \, \mathrm{d}x = \tan(x) + C$$

6.1 The Definite Integral: Antiderivatives

- Since we now know how to differentiate inverse trigonometric functions, for every $s > 0$ we also have:

$$\int \frac{1}{\sqrt{s^2 - x^2}}\,\mathrm{d}x = \arcsin\left(\frac{x}{s}\right) + C$$

$$\int \frac{-1}{\sqrt{s^2 - x^2}}\,\mathrm{d}x = \arccos\left(\frac{x}{s}\right) + C$$

$$\int \frac{1}{s^2 + x^2}\,\mathrm{d}x = \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C$$

6.1 The Definite Integral: Antiderivatives

Example 6.8

Check that $\frac{1}{s}\arctan\left(\frac{x}{s}\right) + C$ is an antiderivative of $\dfrac{1}{s^2 + x^2}$ for $s > 0$.

6.1 The Definite Integral: Antiderivatives

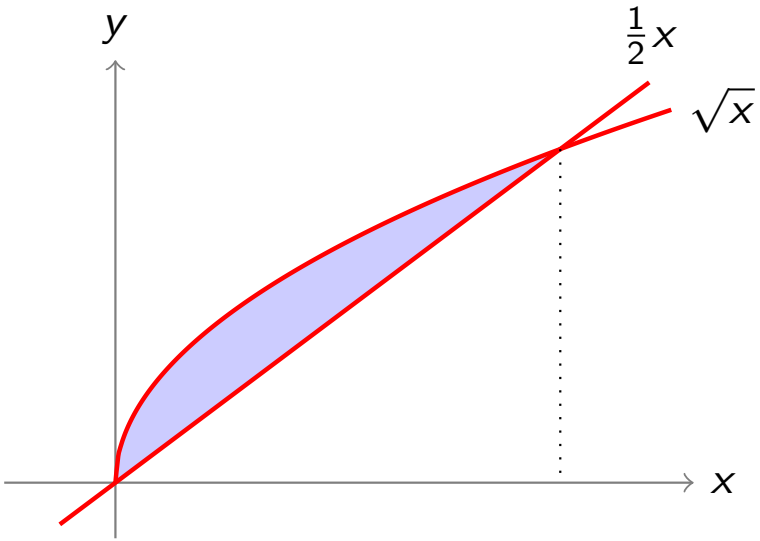
Homework 57

Check the arcsin *and* arccos *antiderivatives are correct.*

6.1 The Definite Integral: Areas Enclosed by Curves

Example 6.9

Find the area of the region enclosed by $y = \sqrt{x}$ and $y = \frac{1}{2}x$.

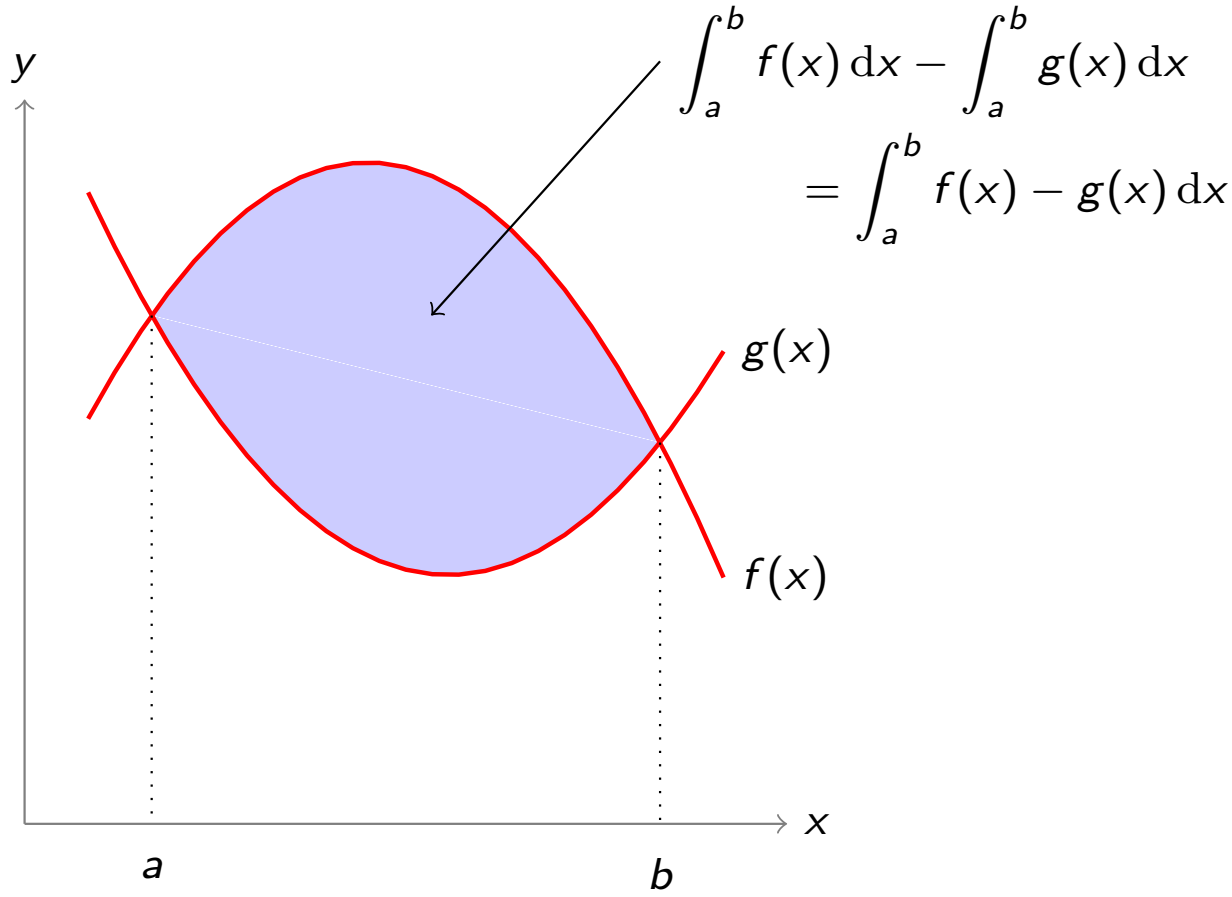


The curves intersect where:

So area =

6.1 The Definite Integral: Areas Enclosed by Curves

- The area enclosed by two graphs can be found by subtracting definite integrals.
- The terminals of these integrals occur where the curves cross.



6.1 The Definite Integral: Areas Enclosed by Curves

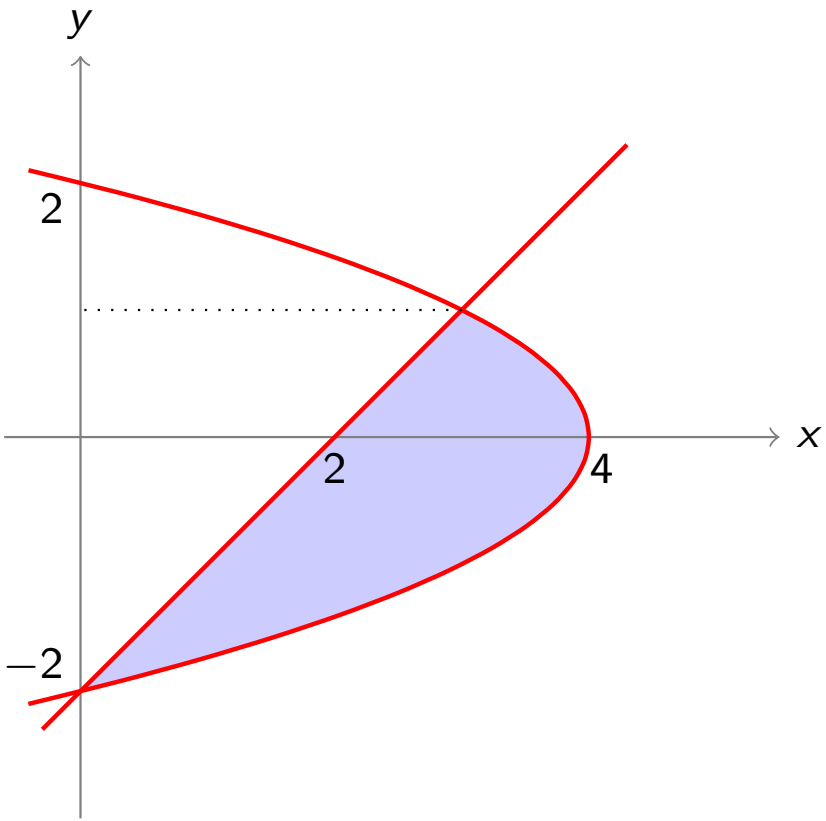
Homework 58

Find the area enclosed by the curves $y = x$ *and* $y = x^3 - x$*. Draw a sketch to help your thinking.*

6.1 The Definite Integral: Areas Enclosed by Curves

Example 6.10

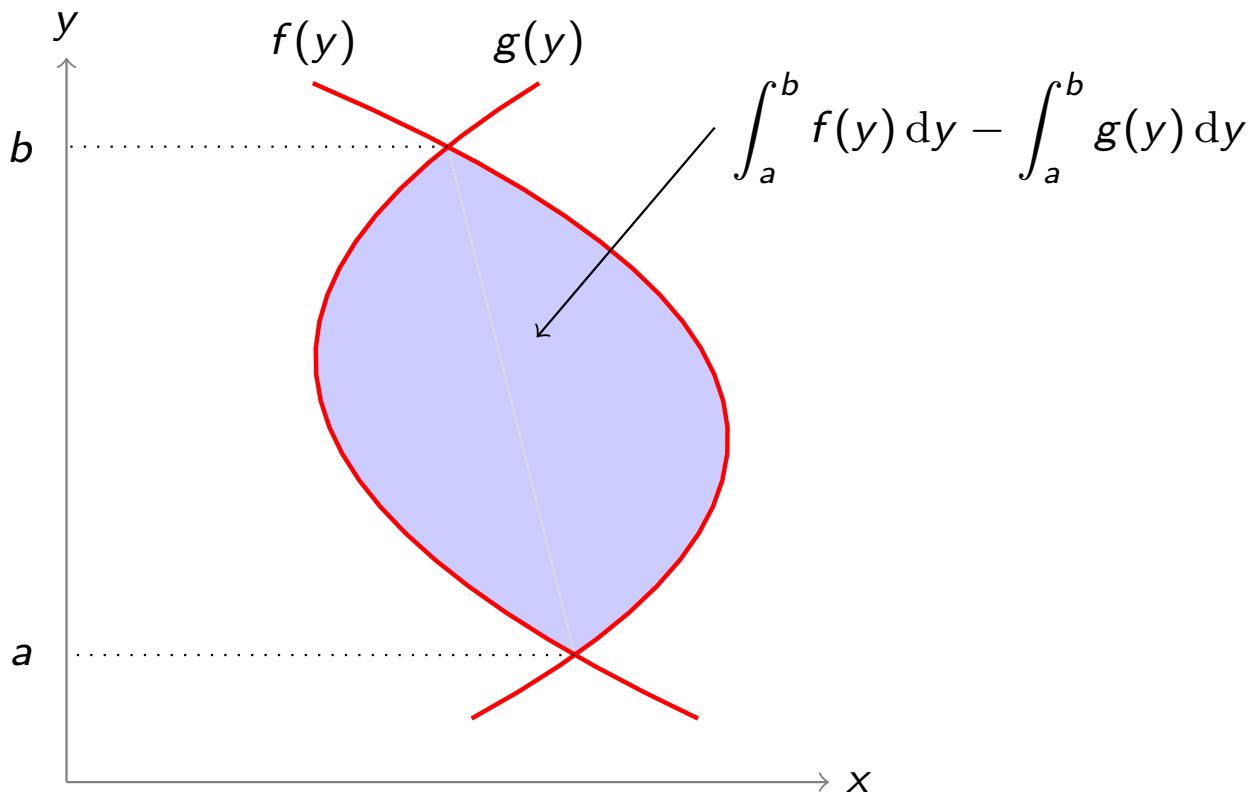
Find the area enclosed by the curves $y = x - 2$ and $x = 4 - y^2$.



- In this case, y is a function of x

6.1 The Definite Integral: Areas Enclosed by Curves

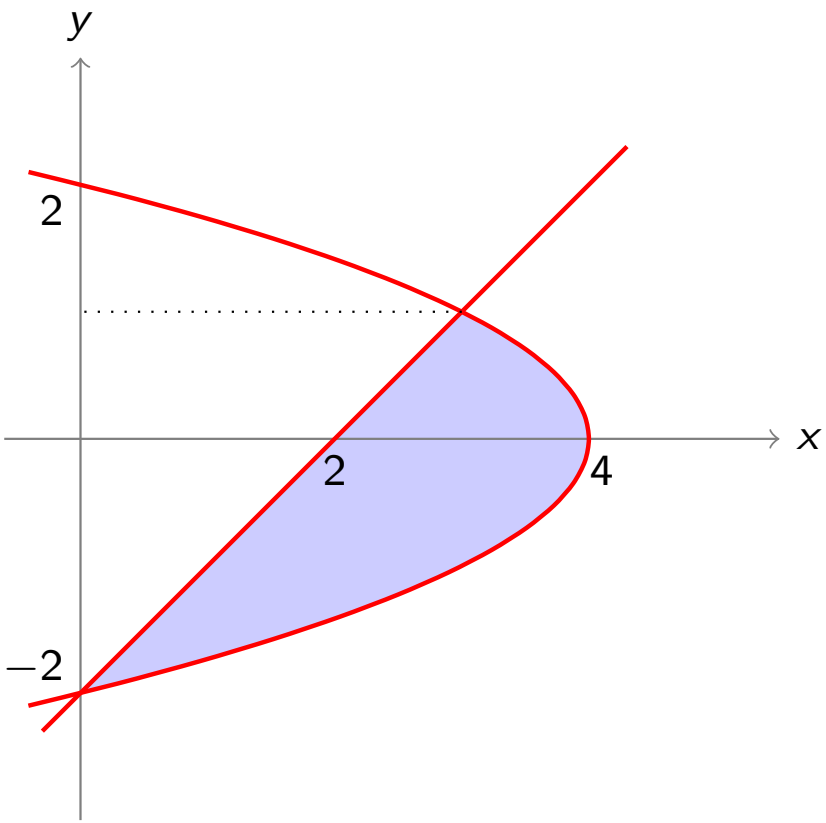
- Suppose we have two relations for which x is a function of y.
- Then we can find the area enclosed by the graphs of the two relations by integrating with respect to y.
- We still need to find where the curves intersect.



6.1 The Definite Integral: Areas Enclosed by Curves

Example 6.11

Find the area enclosed by the curves $y = x - 2$ and $x = 4 - y^2$.



6.1 The Definite Integral: Areas Enclosed by Curves

Homework 59

Find the area enclosed between the curves $y^2 - x = 1$ *and* $y^2 + x = 1$.

6.2 Techniques for Integration

- Unlike differentiation, there are no *nice* techniques for products and quotients.

$$\int f(x)g(x)\,\mathrm{d}x \neq f(x)\int g(x)\,\mathrm{d}x$$

$$\int f(x)g(x)\,\mathrm{d}x \neq \int f(x)\,\mathrm{d}x \int g(x)\,\mathrm{d}x$$

$$\int \frac{f(x)}{g(x)}\,\mathrm{d}x \neq \frac{\int f(x)\,\mathrm{d}x}{\int g(x)\,\mathrm{d}x}$$

$$\int \frac{1}{g(x)}\,\mathrm{d}x \neq \log(|g(x)|) + C$$

6.2 Integration by Substitution

- Consider using the chain rule "backwards".

$$\frac{d}{dx}\left[\sin^5(x)\right] = 5\sin^4(x)\cos(x).$$

Therefore, by the Fundamental Theorem of Calculus:

$$\int_a^b 5\sin^4(x)\cos(x)\,\mathrm{d}x = \int_a^b \frac{d}{dx}[\sin^5(x)]\,\mathrm{d}x = \sin^5(b) - \sin^5(a).$$

6.2 Integration by Substitution

- Notice that our integrand $5\sin^4(x)\cos(x)$ can be written

$$g(h(x))h'(x)$$

  where $g(x) = 5x^4$ and $h(x) = \sin(x)$.
- We can integrate any function of this form provided we know an antiderivative G of g:

$$\begin{aligned}\int_a^b g\big(h(x)\big)h'(x)\,\mathrm{d}x &= \int_a^b G'\big(h(x)\big)h'(x)\,\mathrm{d}x \\ &= \int_a^b \frac{d}{dx}\big[G(h(x))\big]\,\mathrm{d}x \qquad \text{[Chain rule]} \\ &= \big[G(h(x))\big]_a^b = G(h(b)) - G(h(a))\end{aligned}$$

- It is convenient to split this type of integration into two easier steps.

6.2 Integration by Substitution

- We achieve this by working backwards from our previous answer in a slightly different way:

$$\begin{aligned} G(h(b)) - G(h(a)) &= \big[G(x)\big]_{h(a)}^{h(b)} \\ &= \int_{h(a)}^{h(b)} G'(x)\,\mathrm{d}x = \int_{h(a)}^{h(b)} g(x)\,\mathrm{d}x. \end{aligned}$$

- Combining these two expressions for $G(h(b)) - G(h(a))$ yields Step 1, the integration by substitution rule:

Definite Integration by Substitution

$$\int_a^b g\big(h(x)\big)h'(x)\,\mathrm{d}x = \int_{h(a)}^{h(b)} g(x)\,\mathrm{d}x.$$

6.2 Integration by Substitution

Example 6.12

Find $\displaystyle\int_1^{e^{\frac{\pi}{2}}} \frac{\sin(\log(x))}{x}\,\mathrm{d}x.$

- After applying the substitution rule, Step 2 is to integrate the simpler expression $g(x)$.

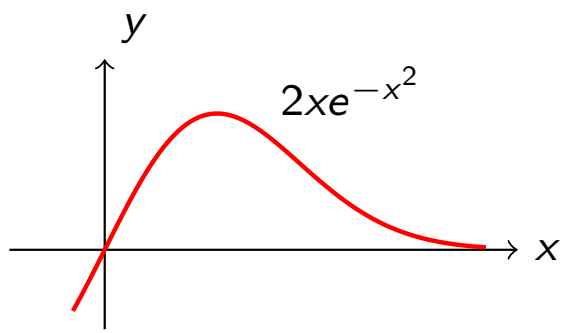
6.2 Integration by Substitution

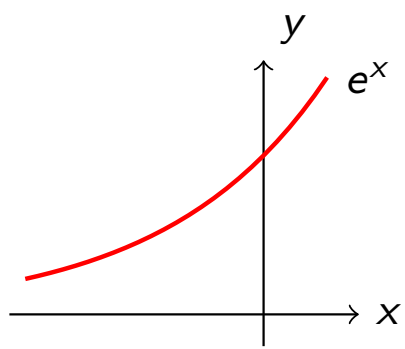
Example 6.13

Use integration by substitution to simplify

$$\int_0^1 2xe^{-x^2}\,\mathrm{d}x$$

and illustrate this in terms of areas on the graphs below.





6.2 Integration by Substitution

Example 6.14

Suppose $h : \mathbb{R} \longrightarrow \mathbb{R}$ is differentiable, $h(4) = 1$ and $h(7) = 0$. Find

$$\int_4^7 (h^2(x) + 4h(x) + 7)h'(x)\,\mathrm{d}x.$$

6.2 Integration by Substitution

Homework 60

Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ *be a function such that* $\int_0^1 f(x)dx = \frac{\pi}{2}$.
Use integration by substitution to evaluate

$$\int_1^e \frac{2f(\log(x))}{x}\, dx.$$

6.2 Integration by Substitution

- Setting $u = h(x)$ gives a version of the substitution rule that helps us to find antiderivatives:

Substitution rule (antiderivative form)

$$\int g(u)\frac{du}{dx}\,\mathrm{d}x = \int g(u)\,\mathrm{d}u.$$

- The challenge is to express the integrand as $g(u)\dfrac{du}{dx}$.

6.2 Integration by Substitution

Example 6.15

Find $\int 2x(x^2 - 5)^4 \, \mathrm{d}x$.

6.2 Integration by Substitution

Example 6.16

Find $\displaystyle\int \cos(3x)\sqrt{\sin(3x)+4}\,\mathrm{d}x$.

6.2 Integration by Substitution

Homework 61

Find $\int (x^4 + 1)\, e^{x^5+5x}\, \mathrm{d}x.$

6.2 Integration by Substitution

Homework 62

Find $\displaystyle\int \frac{x^2+1}{\sqrt{x^3+3x}}\,\mathrm{d}x.$

6.2 Integration by Substitution

Thus far the substitution rule may not appear particularly powerful, but when cleverly applied, it allows us to tackle a surprising range of integration and anti-differentiation problems.

1. Linear substitutions, where we express surprisingly difficult looking integrals in an easier form in cases where the "inside" function $u = mx + c$ is linear. We then to express the remaining x's in terms of u (using the inverse of u).
2. Products of powers of trigonometric functions, where we again use a technique to make difficult looking problems much easier (in cases where at least one of the powers is odd).

We look at technique 1 (and leave technique 2 for tutorial)

6.2 Integration by Substitution: Linear Substitution

Example 6.17

Find $\int (2x+1)\sqrt{x-3}\ \mathrm{d}x$.

Example 6.18

Evaluate

$$\int \frac{4x^2}{\sqrt{1-2x}}\ \mathrm{d}x$$

Example continued

6.3 Rational Functions with Small Denominators

The remainder of our integration discussion will be about how to anti-differentiate rational functions – functions of the form

$$\frac{p(x)}{q(x)}$$

where p and q are polynomials. The first step on this journey is to learn to use the substitution rule to anti-differentiate rational functions for which q is either linear (degree 1) or irreducible quadratic (degree 2) and p is of strictly smaller degree than q.

As it turns out, these are the only kinds of rational functions we need to anti-differentiate. In the remaining part of our journey, we will learn how to break up any rational functions into a polynomial plus a sum of terms of these types.

We first recall how to anti-differentiate $\frac{1}{x}$.

6.3 Rational Functions with Small Denominators: $\frac{1}{x}$

6.3 Rational Functions with Small Denominators: Linear Denominators

- Linear substitution allows us to find antiderivatives of rational functions with denominator of degree one.
- We have just seen one example:

$$\int \frac{1}{x}\,\mathrm{d}x = \log(|x|) + C\ , \qquad x \neq 0.$$

- This lets us approach any integral of the type

$$\int \frac{1}{ax+b}\,\mathrm{d}x$$

by using the substitution $u = ax + b$.

6.3 Rational Functions with Small Denominators: Linear Denominators

Example 6.19

Find $\displaystyle\int \frac{1}{2x-3}\,\mathrm{d}x$ by two methods:

1. Using the substitution $u = 2x - 3$.
2. By first taking out a factor $\frac{1}{2}$.

and explain the apparent difference between the two answers.

Example continued

6.3 Rational Functions with Small Denominators: Irreducible Quadratic Denominators

- For an irreducible quadratic denominator $q(x) = x^2 + bx + c$, we can rewrite the denominator as

$$q(x) = u^2 + s^2$$

  with $s > 0$ by completing the square and making a linear substitution using u.
- This usually splits the integrand into two terms:
  1. One can be integrated by substitution, giving $\log(q(x))$.
  2. The other can be integrated using the antiderivative

$$\int \frac{1}{s^2 + u^2} \, \mathrm{d}u = \frac{1}{s} \arctan\left(\frac{u}{s}\right) + C.$$

- But how can we be sure that $q(x) = u^2 + s^2$ with $s > 0$?

6.3 Rational Functions with Small Denominators: Completing the Square

- A quadratic $p(x) = x^2 + bx + c$ can always be expressed as a perfect square plus a constant by <u>completing the square</u>:

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + r$$

where $r = c - \frac{b^2}{4}$.

Example 6.20

Show that if $p(x)$ has no real roots, the remainder r is positive.

6.3 Rational Functions with Small Denominators

Example 6.21

Find $\displaystyle\int \frac{1}{x^2+2x+2}\,\mathrm{d}x$

6.3 Rational Functions with Small Denominators

Homework 63

Find $\displaystyle\int \frac{6}{x^2 + 2x + 5}\,\mathrm{d}x$

6.3 Rational Functions with Small Denominators

Example 6.22

Find $\displaystyle\int \frac{x+1}{x^2+2x+2}\,\mathrm{d}x$

6.3 Rational Functions with Small Denominators

6.3 Rational Functions with Small Denominators: Linear Numerator

Example 6.23

Find $\displaystyle\int \frac{x-2}{x^2+2x+2}\,\mathrm{d}x$

6.3 Rational Functions with Small Denominators: Linear Numerator

6.3 Rational Functions with Small Denominators

Homework 64

Find $\displaystyle\int \frac{3x+5}{x^2-4x+5}\,\mathrm{d}x$

6.3 Rational Functions with Small Denominators

Example 6.24

Evaluate

$$\int \frac{1}{2x^2 + 2x + 4} \, \mathrm{d}x.$$

6.4 Rational Functions: Partial Fractions

So far we have only considered rational functions whose denominators were either linear or quadratic.
The key to integrating more complicated rational functions is to decompose them into a sum of simpler fractions that we already know how to integrate. Consider, for example:

$$\frac{6x^4 - 6x^3 - 4x^2 + 12x - 14}{(x-1)^2(x+2)(x^2+1)} = \frac{3}{x-1} - \frac{1}{(x-1)^2} + \frac{2}{x+2} + \frac{x-4}{x^2+1}$$

In the previous section, we learned how to integrate each of the terms on the RHS of this example. You can check that this equality holds by putting the RHS over a common denominator.

Question: How do we find this decomposition?

6.4 Rational Functions: Partial Fractions

When integrating a rational function $f(x) = \dfrac{p(x)}{q(x)}$:

- We need only to know how to handle the case where the degree of p is strictly smaller than the degree of q.
- (If not, polynomial long division gives a polynomial part plus a remainder that satisfies this requirement.)
- The next step is to factor q into linear and irreducible quadratic factors with real coefficients (if we can):

$$(x - a_1)\dots(x - a_j)(x^2 + b_1 x + c_1)\dots(x^2 + b_k x + c_k).$$

- The <u>Partial Fractions Theorem</u> (PFT) then tells us how to break $f(x)$ into pieces we (mostly) know how to integrate.

6.4 Rational Functions: Partial Fractions

Theorem 6.25 (Partial Fractions with Distinct Factors)

Provided $\deg(p) < \deg(q)$ *and the factors of* $q(x)$ *are distinct:*

$$\frac{p(x)}{(x-a_1)\dots(x-a_j)(x^2+b_1x+c_1)\dots(x^2+b_kx+c_k)}$$

can always be written as a sum

$$\frac{A_1}{x-a_1}+\dots+\frac{A_j}{x-a_j}+\frac{B_1x+C_1}{x^2+b_1x+c_1}+\dots+\frac{B_kx+C_k}{x^2+b_kx+c_k}$$

where $A_1,\dots,A_j$, $B_1,\dots,B_k$, $C_1,\dots,C_k$ *are real constants.*

- Each linear factor $x-a$ gives a term $\dfrac{A}{x-a}$.
- Each irreducible x^2+bx+c gives a term $\dfrac{Bx+C}{x^2+bx+c}$.
- Our task is to find the A_i's, B_i's and C_i's.

6.4 Rational Functions: Partial Fractions

Example 6.26

Decompose $f(x) = \dfrac{9x+1}{(x-3)(x+1)}$ into partial fractions and use the decomposition to find

$$\int f(x)\,\mathrm{d}x.$$

Example continued

6.4 Rational Functions: Partial Fractions

Homework 65

Find $\displaystyle\int \frac{7}{x^2 + 3x - 10}\,\mathrm{d}x$

6.4 Rational Functions: Partial Fractions

Example 6.27

Find $\displaystyle\int \frac{3x^2 - 2x + 1}{(x+1)(x^2+2x+2)}\,\mathrm{d}x$

Example continued

6.4 Rational Functions: Partial Fractions

- Recall that each distinct factor in the denominator gives a term in the partial fraction decomposition as follows:

$$(x-a) \quad \text{gives} \quad \frac{A}{x-a}$$

$$(x^2+bx+c) \quad \text{gives} \quad \frac{Bx+C}{x^2+bx+c}$$

- When factors are repeated:

$$(x-a)^k \quad \text{gives} \quad \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}$$

$$(x^2+bx+c)^k \quad \text{gives} \quad \frac{B_1x+C_1}{x^2+bx+c} + \frac{B_2x+C_2}{(x^2+bx+c)^2} + \cdots + \frac{B_kx+C_k}{(x^2+bx+c)^k}$$

6.4 Rational Functions: Partial Fractions

- Suppose partial fractions gives a term

$$\frac{Ax + B}{(x^2 + ax + b)^n}.$$

- We can still write the denominator in completed square form $(u^2 + s^2)^n$, giving:

$$\frac{pu + q}{(u^2 + s^2)^n} = \frac{pu}{(u^2 + s^2)^n} + \frac{q}{(u^2 + s^2)^n}$$

- We can anti-differentiate the first term by substitution
- But we have not learned a technique for terms of the form:

$$\frac{q}{(u^2 + s^2)^n} = \frac{q}{(x^2 + ax + b)^n} \qquad [n > 1].$$

- Such terms can be integrated using MAST10006 techniques.

Example 6.28

Find $\displaystyle\int \frac{3x+1}{x^2+4x+4}\,\mathrm{d}x$

Example continued

6.4 Rational Functions: Partial Fractions

Example 6.29

Solve Example 6.28 using the linear substitution $u = x + 2$.

Homework 66

Find $\displaystyle\int \frac{2x-1}{x^2-6x+9}\,\mathrm{d}x$

6.4 Rational Functions: Partial Fractions

Example 6.30

Find $\int f(x)\,\mathrm{d}x$ where $f(x) = \dfrac{2x^3 - 3x^2 - 8x + 24}{x^2 - 4}$.

Example continued

6.4 Rational Functions

Homework 67

Write down the partial fractions decomposition you would use for each of the following:

(a) $\dfrac{x+2}{x^2+4x-5}$

(b) $\dfrac{1-2x}{x^2+6x+9}$

(c) $\dfrac{3}{x^2-4}$

6.4 Rational functions: Integration of Rational Functions - Overview

- Perform polynomial long division so that the numerator of the remainder has strictly smaller degree than the denominator (if necessary)
- Factor the denominator into linear and quadratic factors.
- Translate the factorisation into the appropriate form of the partial fraction decomposition.
- Solve to find the constants.
- Integrate each term of the partial fraction decomposition separately and put the results together.

6.4 Rational functions: Integration of Rational Functions - Overview

- When anti-differentiating terms of the form

$$\frac{p(x)}{q(x)} = \frac{linear}{quadratic}$$

the method we use depends on whether q is irreducible:

1. If q is irreducible, we write $q(x) = u^2 + s^2$ by completing the square and make a linear substitution using u.
2. If $q(x) = (x - a)(x - b)$, we use partial fractions